

Javier Parada

Cybersecurity Researcher & Software Engineer

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Cybersecurity Researcher and PhD Candidate focused on *Cyber Threat Intelligence* and *Adversarial Emulation* — simulating attacker behaviour to anticipate threats and strengthen defenses. Former Software Engineer with experience designing scalable, high-availability, event-driven systems.

Experience

Eurecat — Technology Centre

Barcelona, Spain

Cybersecurity Researcher

2024 – Present

- Cybersecurity research focused on Cyber Threat Intelligence, Threat Hunting and Adversarial Emulation. Analysis and detection of advanced threats, simulation of malicious behaviours, and development of tools and methodologies to anticipate and respond to security incidents.

NICT — Japan National Cybersecurity Agency

Tokyo, Japan

Cybersecurity Researcher

2026 · 4 months

- Cybersecurity research focused on Cyber Threat Intelligence, Threat Hunting and Adversarial Emulation. Analysis and detection of advanced threats, simulation of malicious behaviours, and development of tools and methodologies to anticipate and respond to security incidents.

NICS Lab — Research Group

Málaga, Spain

Cybersecurity Researcher

2023 – Present

- Research on Cyber Threat Intelligence and Adversary Emulation applied to Critical Infrastructures. Leveraging digital twins to model and simulate complex systems for threat anticipation, and deploying honeypots to attract, deceive and analyse adversarial behaviours.

T2C — IT Consulting Company

Barcelona, Spain

Software Engineer

2020 – 2023

- Design and development of systems using Domain-Driven Design, Microservices and Hexagonal Architecture, built for high scalability and large data volumes, with emphasis on reliability, observability and event-driven communication.

DeveryWare — IT Consulting Company

Paris, France

Software Developer

2019 – 2020

- Application development with Bash, Linux, Angular, JavaScript, MongoDB, Docker and Node.js, applying SOLID principles for clean, scalable and maintainable code.

Internalia Group — Software Product Company

Málaga, Spain

Software Developer

2018

- Web application development with PHP, JavaScript, SQL and Linux systems, applying SOLID principles for clean, scalable and maintainable code.

Education

University of Málaga

Málaga, Spain

Ph.D. in Computer Science

May 2024

- Research on Cyber Threat Intelligence, Threat Hunting and Adversarial Emulation: analysis and detection of advanced threats, simulation of malicious behaviours, and tools and methodologies to anticipate and respond to security incidents in complex environments.
- Skills: Cybersecurity · Artificial Intelligence · Cyber Threat Intelligence · High Availability

University of Málaga

Master's in Computer Science

Málaga, Spain

Oct 2021 – Mar 2024

- Specialisation in Cybersecurity: pentesting, malware analysis, secure programming, vulnerability discovery and exploitation, and system and data privacy.
- Thesis: *TanukiPot: Honeypot based on Digital Twins for Critical Infrastructures*.
- Skills: Cybersecurity · Pentesting · Malware Analysis · Industrial Infrastructures

University of Málaga

Bachelor's in Computer Science

Málaga, Spain

Sep 2014 – Sep 2020

- Specialisation in Software Engineering: data structures, algorithm analysis and design, software development, databases and cybersecurity.
- Final Degree Project: *Emergency Control and Prevention Platform*.
- Skills: Computer Science · Network Design · Cybersecurity · Programming · Project Management

Publications

Adaptive Adversary Emulation Framework for Cyberwarfare Operations

IEEE ComMag

IEEE Communications Magazine

2026

- OPFOR (Operational Platform for Offensive Replication), a modular framework for Adversary Emulation based on behaviours attributable to real threat actors, integrated within the Cyber Threat Hunting paradigm to protect Critical Infrastructures. It formalises Adaptive Adversary Emulation (AAE), combining CTI with an internal understanding of network topology weaknesses to generate infrastructure-aware attack paths. A Stuxnet-inspired operation, executed by a Reinforcement Learning agent on a simulated nuclear plant network, validates the approach and informs defensive prioritisation and early detection strategies.
- Authors: J. Parada, C. Alcaraz, M. Reyes, J. Lopez.

Digital Twin for Adaptive Adversary Emulation in IIoT Control Networks

ESORICS 2025

European Symposium on Research in Computer Security

Sep 2025

- A framework that uses Digital Twins to emulate adaptive cyber adversaries in IIoT control networks. Reinforcement-learning attackers interact with a virtual replica of the industrial system and adapt their strategies to exploit critical nodes, producing realistic attack scenarios without impacting real infrastructure and enabling proactive evaluation of defensive mechanisms.
- Authors: J. Parada, C. Alcaraz, J. Lopez, J. Caubet, R. Román.
- DOI: [10.1007/978-3-032-07894-0_22](https://doi.org/10.1007/978-3-032-07894-0_22)

Adaptive Adversary Emulation Applied to Critical Infrastructures and IIoT

JNIC 2026

Networks

Jornadas Nacionales de Investigación en Ciberseguridad

May 2026

- An Adversary Emulation solution that dynamically adapts attack tactics to the targeted infrastructure, balancing faithful imitation of the emulated adversary with adaptability to the environment. Guided by machine learning and industrial simulation, an AI agent learns from the environment and its weaknesses, supporting threat hunting and proactive defense evaluation in critical infrastructures and IIoT.
- Proceedings: [JNIC 2026 conference proceedings](#)

Awards

Outstanding Master's Thesis Award

Málaga, Spain

RENIC Cybersecurity Awards — Best National Master's Thesis

- Awarded by the Red de Excelencia Nacional de Investigación en Ciberseguridad (RENIC) for the best national Master's thesis in cybersecurity in Spain.

Certifications

Japanese-Language Proficiency Test — N3 · The Japan Foundation, Málaga

Jan 2025

University Extension Course on Blockchain Technologies · University of Málaga

Jul 2025